Govind Krishnan J

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EDUCATION

B.Tech in Electronics and Communication Engineering **2023–2027**
SCMS School of Engineering and Technology, Angamaly (CGPA - 7.1/10)

Senior Secondary Education (CBSE - Computer Science) **2021–23**
SDV English Medium Higher Secondary School, Alappuzha (80%)

SKILLS

- **Programming Languages:** Python
- **Web Technologies:** Flask, HTML, CSS
- **Databases:** SQLite, SQLAlchemy
- **Libraries and Frameworks:** BeautifulSoup, Requests, NumPy, Scikit-learn, Pandas
- **Developer Tools:** Git, GitHub, VS Code
- **Core Concepts:** Web Scraping, Machine Learning, REST APIs, Authentication Systems, Database Design

PROJECTS

Amazon Price Tracker **2026**
GitHub | Live Demo
Tools used: Python, Flask, SQLite, SQLAlchemy, BeautifulSoup, ScrapingAnt API, Scikit-learn, NumPy, Chart.js

- Developed a full stack web application that tracks Amazon India product prices via automated scraping, with real time monitoring and automated email alerts (Flask-Mail, Gmail SMTP) when prices hit targets.
- Built a machine learning prediction module using Linear Regression to forecast price trends, visualized through interactive Chart.js dashboards.
- Developed secure authentication workflows including OTP based email verification and password hashing using Flask-Login and Flask-Bcrypt, with automated periodic price checks via cronjob.org.

AI Research Agent **2026**
GitHub | Live Demo
Tools used: Python, Groq API, DuckDuckGo Search, BeautifulSoup, Requests

- Developed an autonomous AI research agent that performs end to end topic investigation by searching the web and generating structured research reports through an LLM tool calling workflow.
- Integrated DuckDuckGo Search and web scraping pipelines using Requests and BeautifulSoup to collect and synthesize information from diverse online sources.
- Built an automated citation and report generation system that compiles research findings into structured, exportable documents with referenced sources.

HELIOS - Autonomous Solar Tracker **2025**
Tools used: NodeMCU ESP8266, MPU-6050, QMC5883L Magnetometer, ACS712, LDR Array, IoT

- Designed and built a dual axis autonomous solar tracking system using a 4 LDR array, MPU-6050, and QMC5883L magnetometer for precise orientation sensing and servo control.
- Implemented real time power monitoring using ACS712 and voltage sensors, with an OLED display for live system status and maintenance alerts.
- Developed WiFi/IoT connectivity on NodeMCU ESP8266 with Weather API integration and NTP time synchronization for intelligent, context aware tracking decisions.

CERTIFICATIONS

- Introduction to Machine Learning – NPTEL (IIT Madras) *(certificate)*
- 100 Days of Code: The Complete Python Pro Bootcamp – Udemy *(certificate)*
- Claude 101 – Anthropic *(certificate)*
- Claude Code 101 – Anthropic *(certificate)*